

Camera surveillance

Students Felipe Ultramar, Gustavo Valentim, Isabelle Mamprim, João Pedro Martinez, Lucas Muniz, Pedro Rossetto and Professor Rigel P. Fernandes 

Abstract—In this paper, we present a computer vision approach for the processing and enhancement of document images. The main goal is to improve the readability of digitized or photographed documents, ensuring that they can be accurately read and processed in the future.

Keywords—Computer vision, image treatment, digitalized documents.

I. INTRODUCTION

With the growth of digitized documents, it is essential to ensure that captured images maintain the necessary readability for future processing. However, issues such as low-quality camera images, poor lighting, and the presence of noise can compromise the clarity of the information in documents. In this context, computer vision emerges as a powerful tool for image processing and enhancement. Through pre-processing techniques such as the use of filters and edge detection algorithms, it is possible to improve the visual quality of documents, making them easier to read and store.

II. SYSTEM DESCRIPTION

The developed system aims at analyzing and processing images through Computer Vision techniques. The executed steps follow a structured flow, starting with the import of necessary libraries, such as OpenCV for image processing, NumPy for matrix operations, and Matplotlib for visualization of results. Next, image reading is performed, which is initially loaded in color and subsequently converted to grayscale, facilitating processing in the following steps. In the preprocessing stage, a Gaussian filter is applied to smooth the image and reduce noise, promoting a cleaner base for subsequent analyses. In the edge detection phase, different algorithms are tested, with Roberts appearing to be the most appropriate due to its ability to identify contours with good precision and low computational cost. This edge detector is able to highlight abrupt intensity transitions in the image, which usually correspond to boundaries between different regions of interest. Soon after, segmentation takes place, where the image is binarized through thresholding and then morphological operations, such as closing, are applied to correct flaws and join nearby regions. This flow allows for precise visual and quantitative analysis of the elements present in the image, demonstrating the efficiency of the techniques used in the system.

III. PROPOSED METHOD

To improve the quality of documents and ensure their legibility, this work proposes the use of computer vision techniques focused on image pre-processing. Among the techniques used are smoothing filters, edge detection algorithms,



Fig. 1
ENTER CAPTION

and morphological operations. These techniques allow for noise reduction, enhancement of relevant regions, and correction of imperfections that hinder readability. The goal is to improve the clarity of the image content, facilitating reading for the use of automatic recognition systems.

IV. EXPECTED RESULTS

It is expected that the application of image processing techniques proposed in this work will result in a significant improvement in the legibility of scanned or photographed documents. With the use of filters, an efficient reduction of visual noise is expected. Edge detection should allow better highlighting of regions of interest, especially textual characters. Morphological operations should contribute to image refinement, correcting flaws and removing small imperfections. As a final result, the processed images should present greater contrast between text and background, more defined lines, and absence of unwanted elements, enabling clearer reading by users and recognition systems.

V. CONCLUSIONS

This project made it possible to apply, in practice, fundamental techniques studied in the Computer Vision course for processing document images, such as the use of filters, edge detection, and morphological operations, reinforcing the importance of image pre-processing in applications like digitization. It also provided a clear understanding of how simple

methods can yield significant results in solving real-world problems.

[1]

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